

PHASE LOCKED LOOP (PLL)

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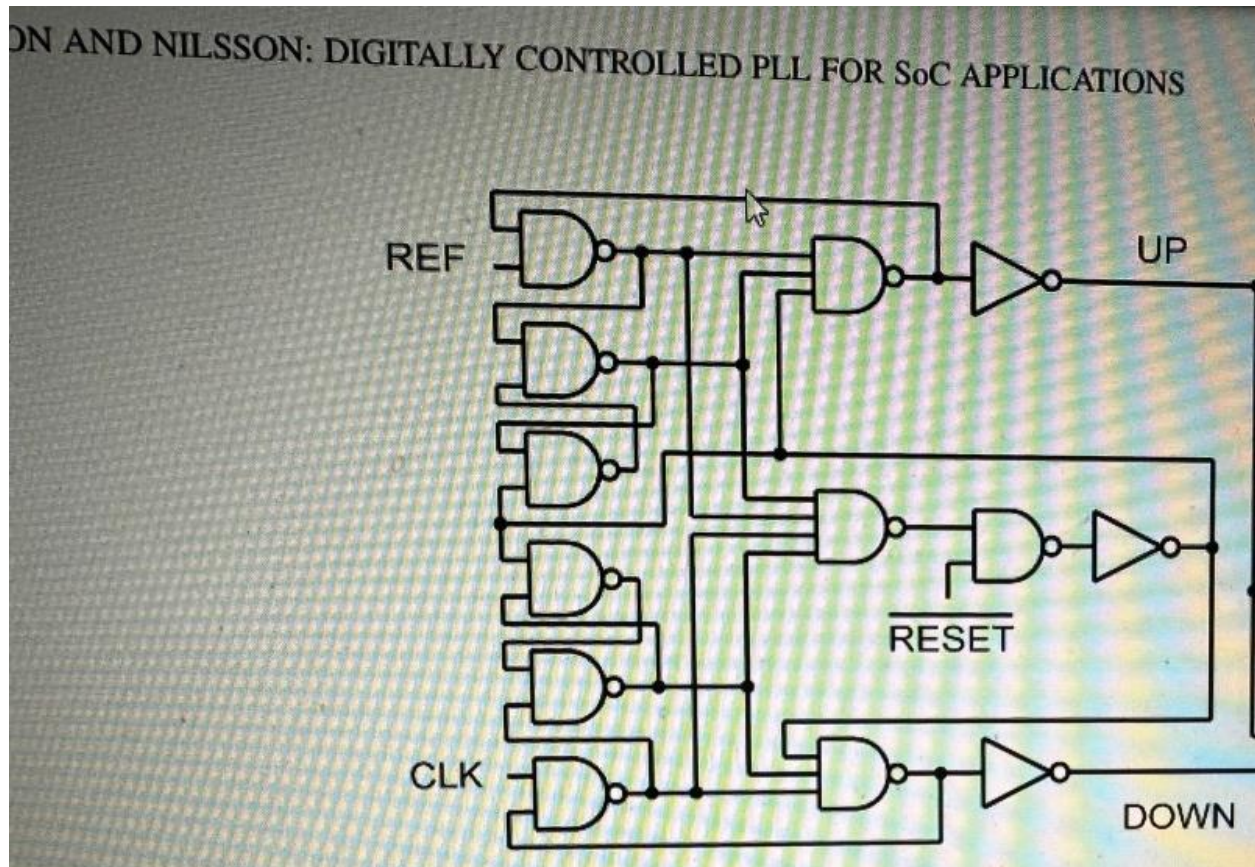
Ahmed sameh 55-16860

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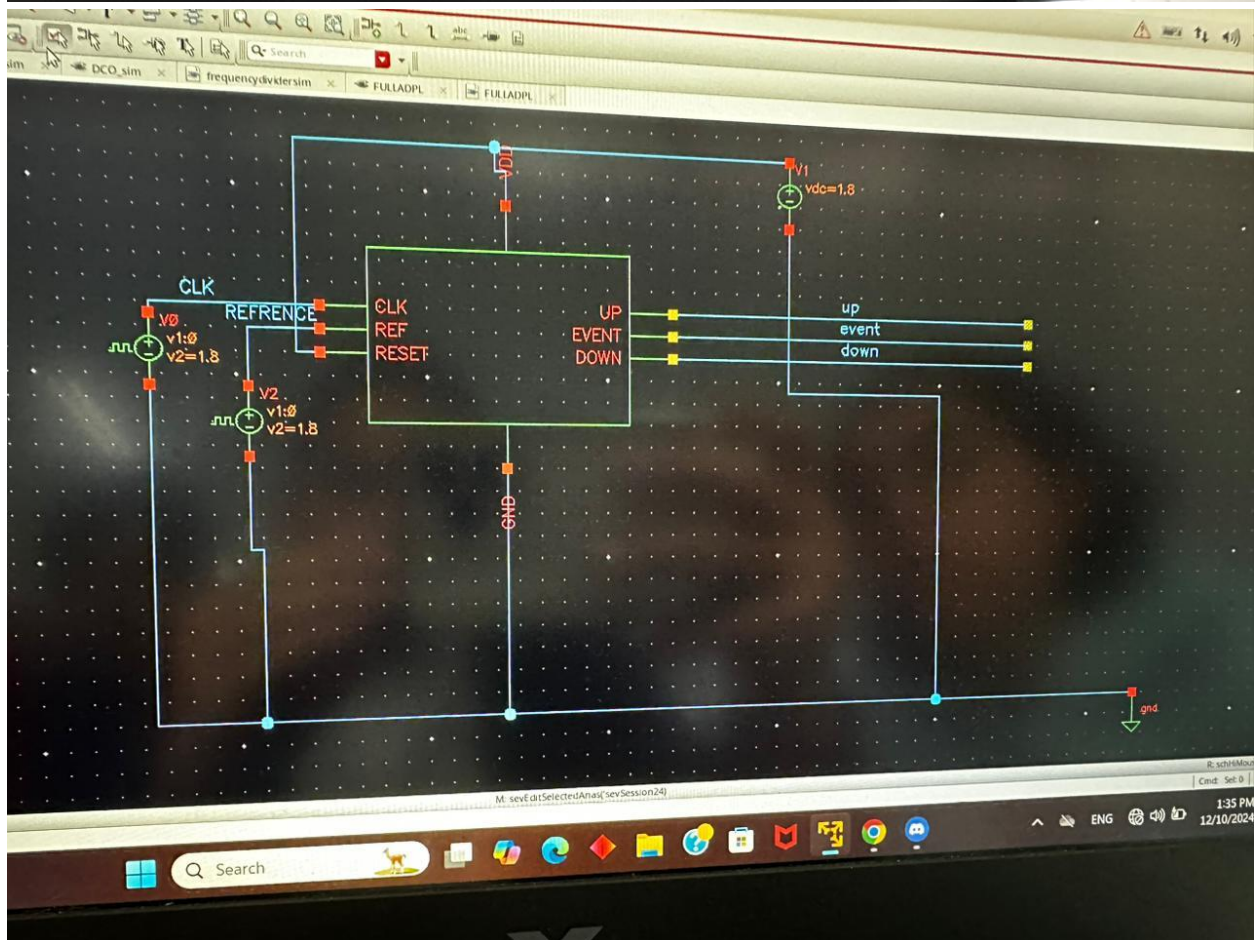
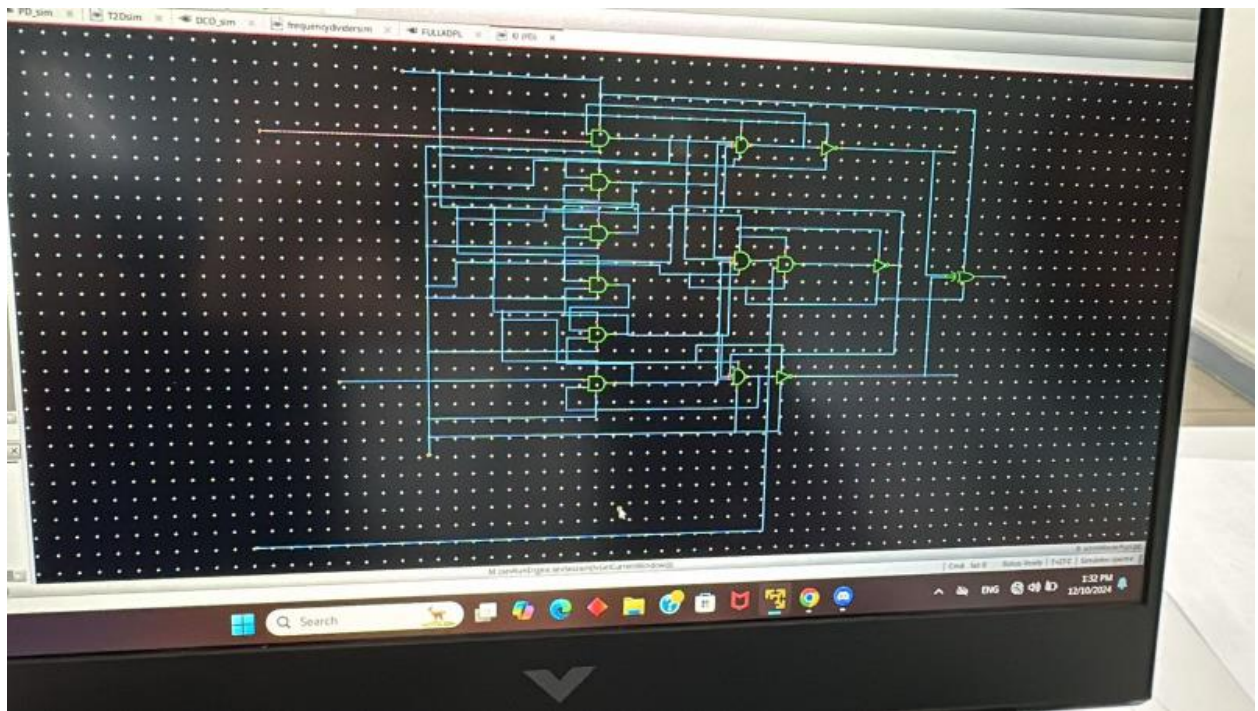
Firstly, using MOSFET Transistors we implemented logic gates,

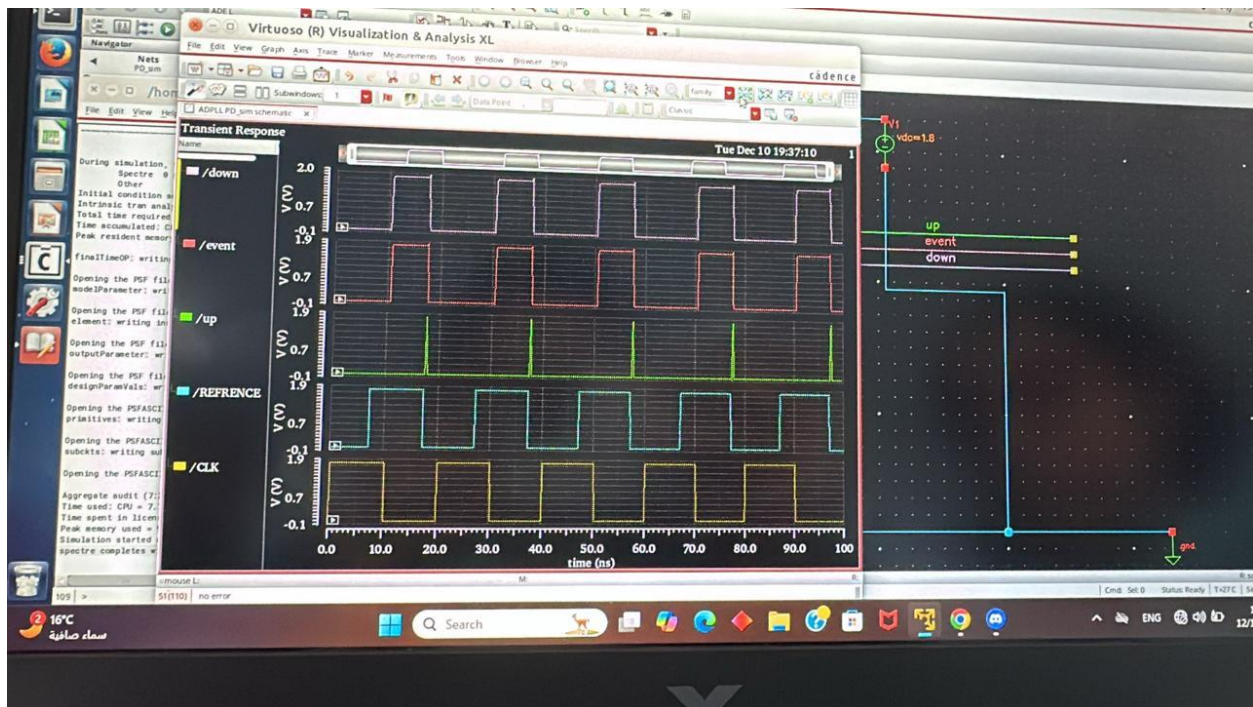
I.PD:

Using the circuit in the figure below



Implementing the circuit in Cadence we have 2 inputs (ref and clk), the desired signal is the reference signal and clk is the signal coming from the frequency divider, the output of the PD is the UP Signal , down signal , and the event . the Up signal represents the ref signal is leading the clk , the down signal represents the ref signal is lagging the clk , the event represents the xoring up and down signal and we have simulated the PD separately in the figure below.

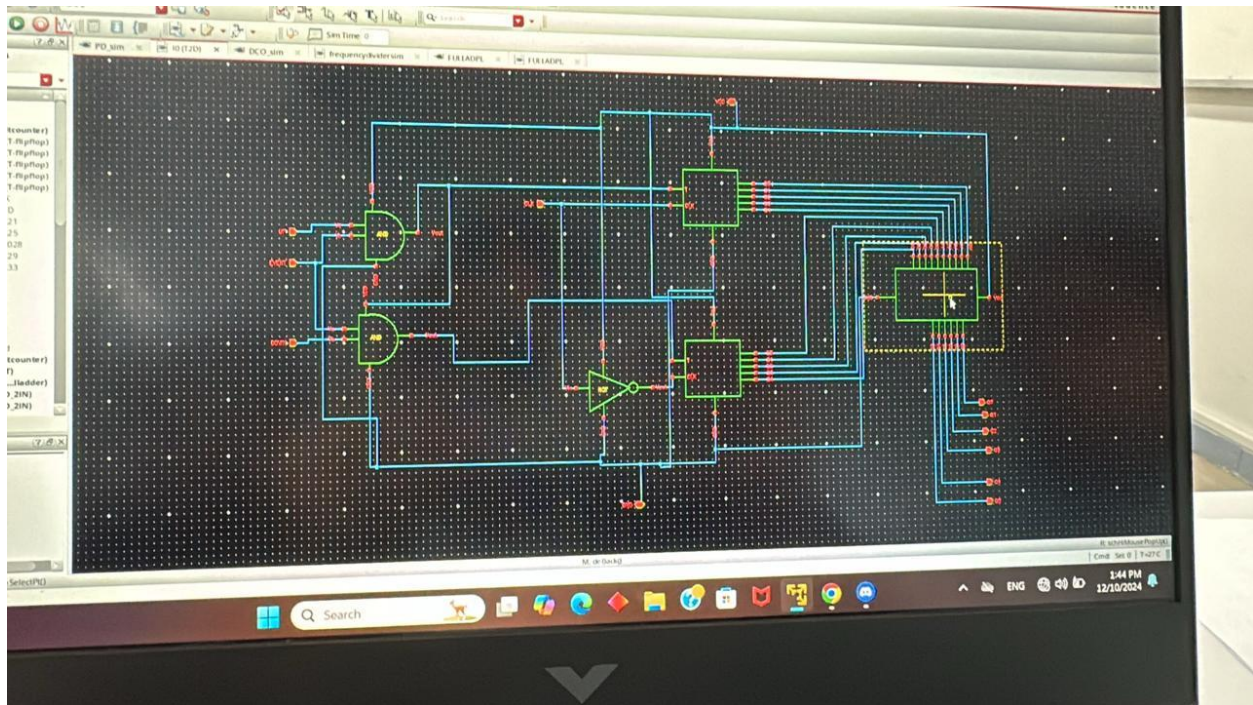




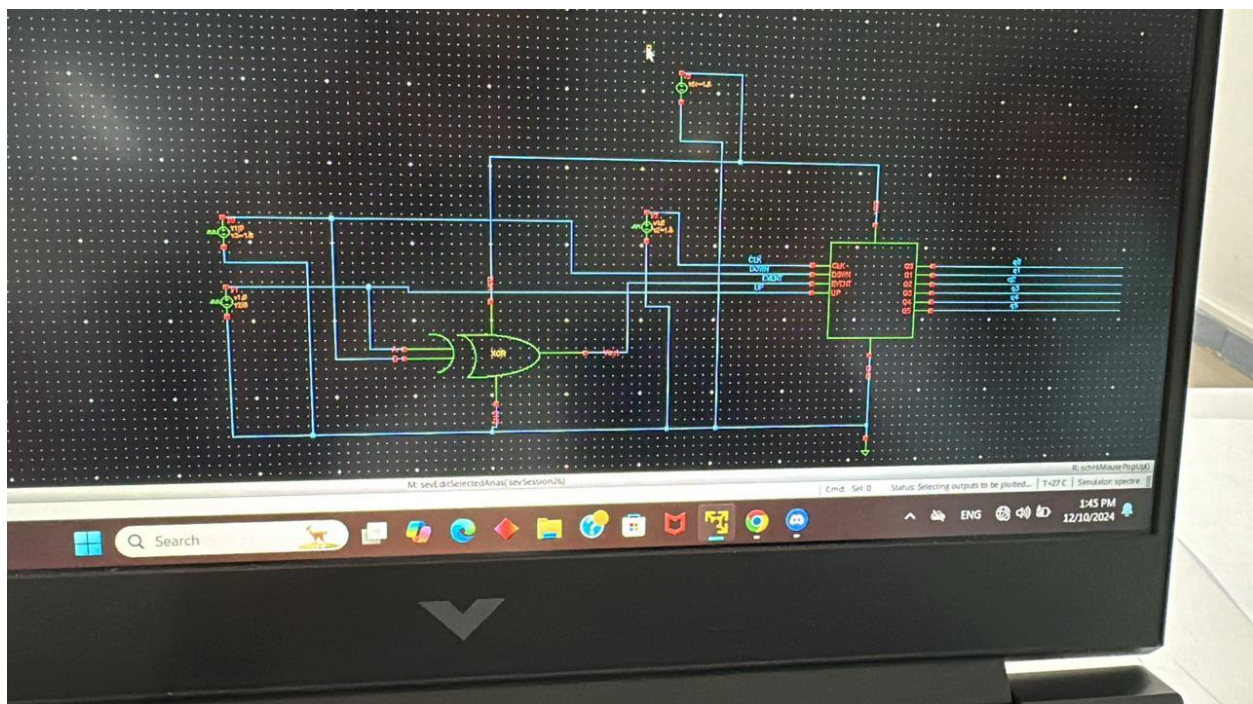
II.T2D AND LOOP FILTER:

From the PD by anding the even and the up signals and entering them to the up counter we implemented the counter using 5 t-Flip flops to get 5 bit output and second input that came from output from the DCO. After this, the down counter is anding the down and the event and entering them to 5 bit counter with notting high frequency that comed from the DCO.

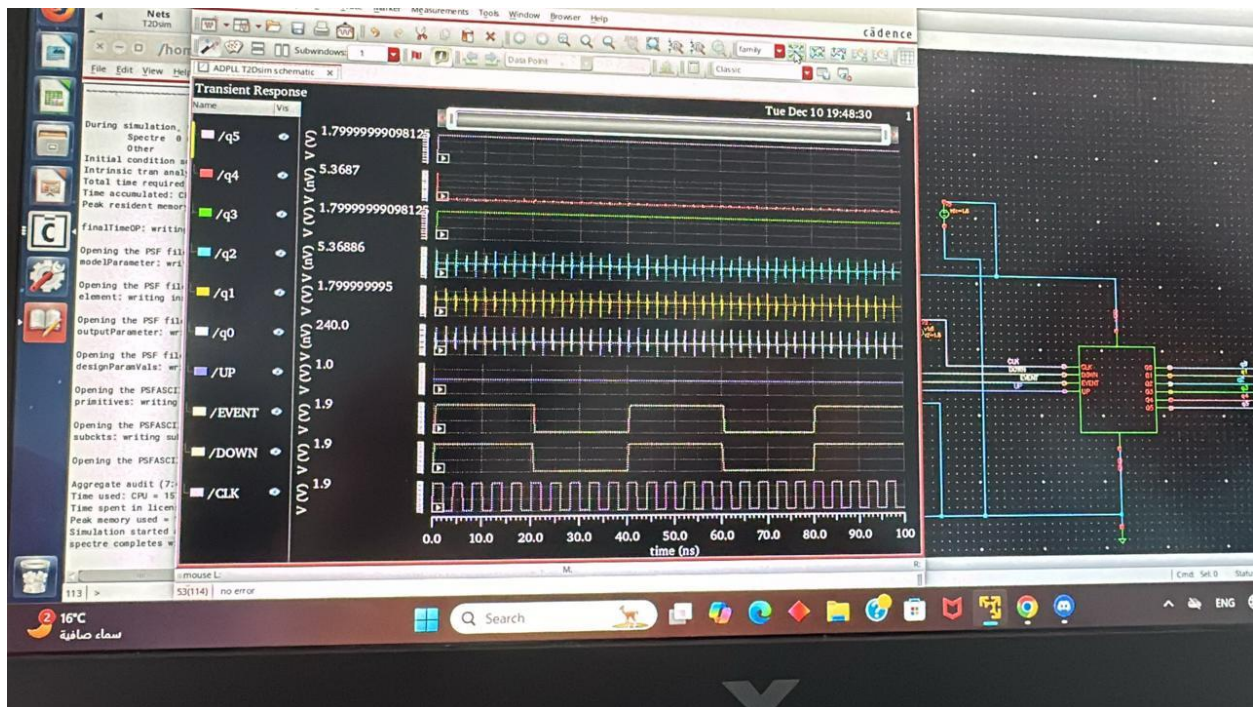
After that, we will add them by full bit adder will give us 5 bit output and 1 carrying bit



And we simulated it separately in the figure below:

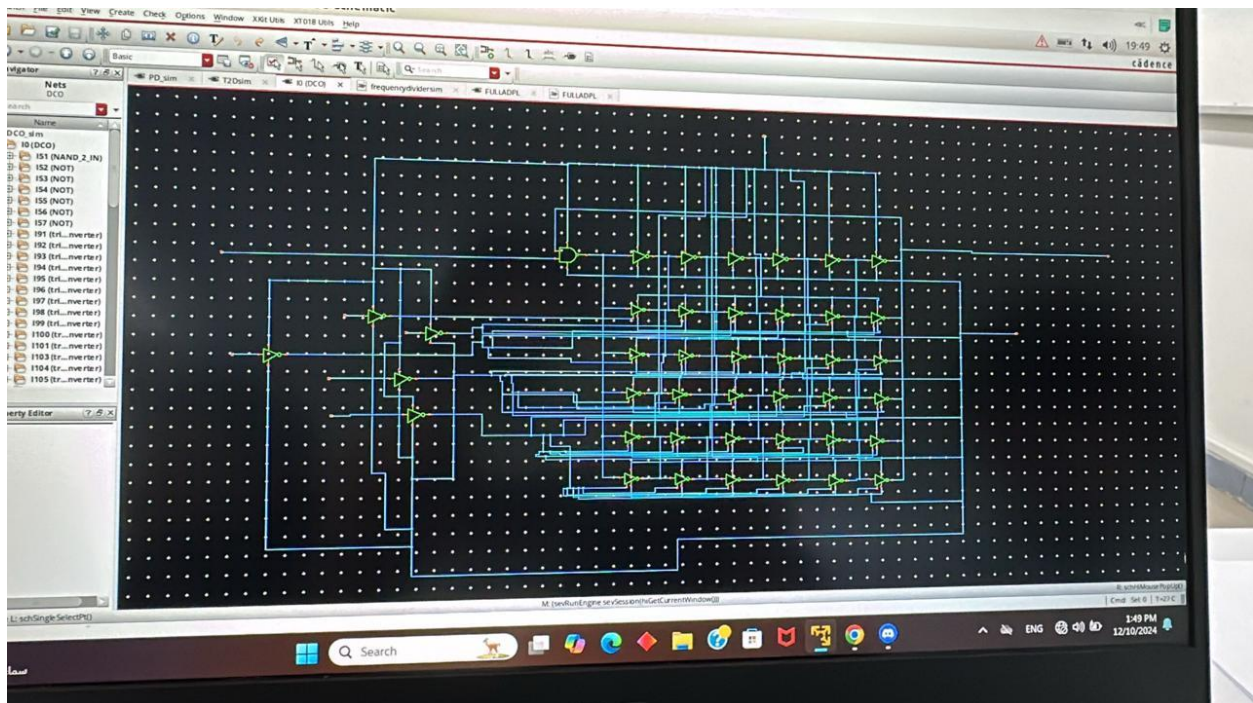


And the results in the figure below:

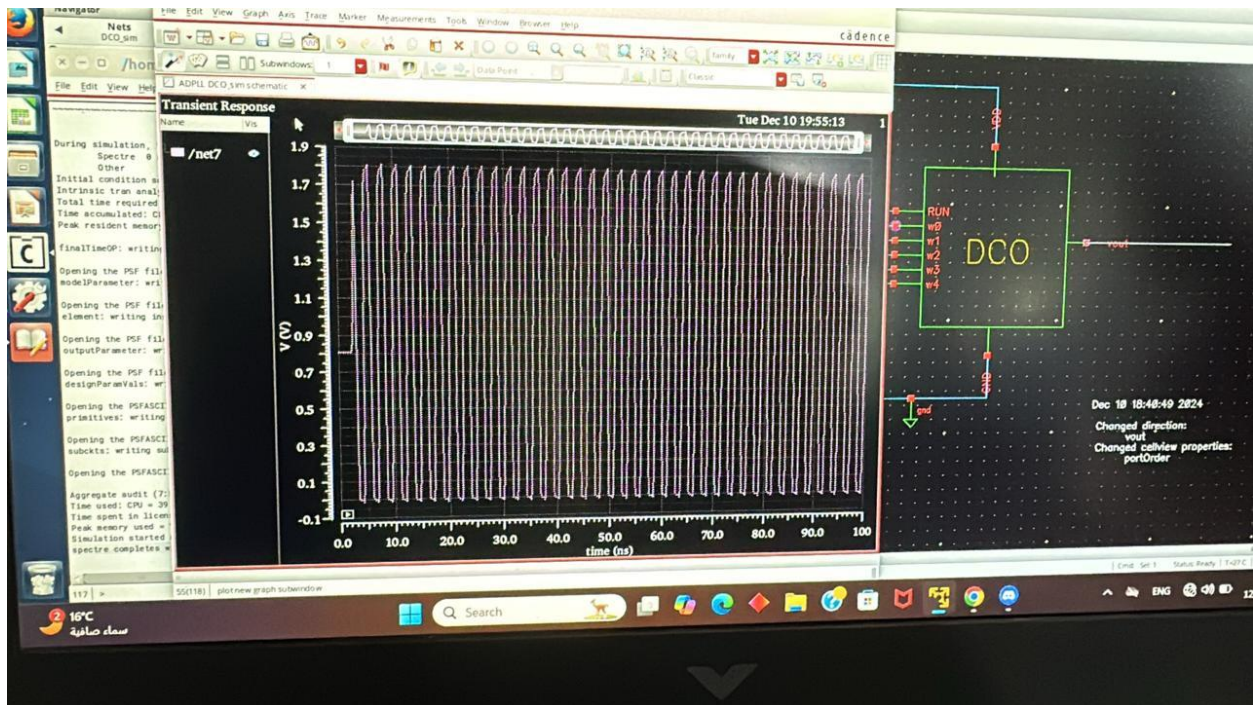


III.DCO:

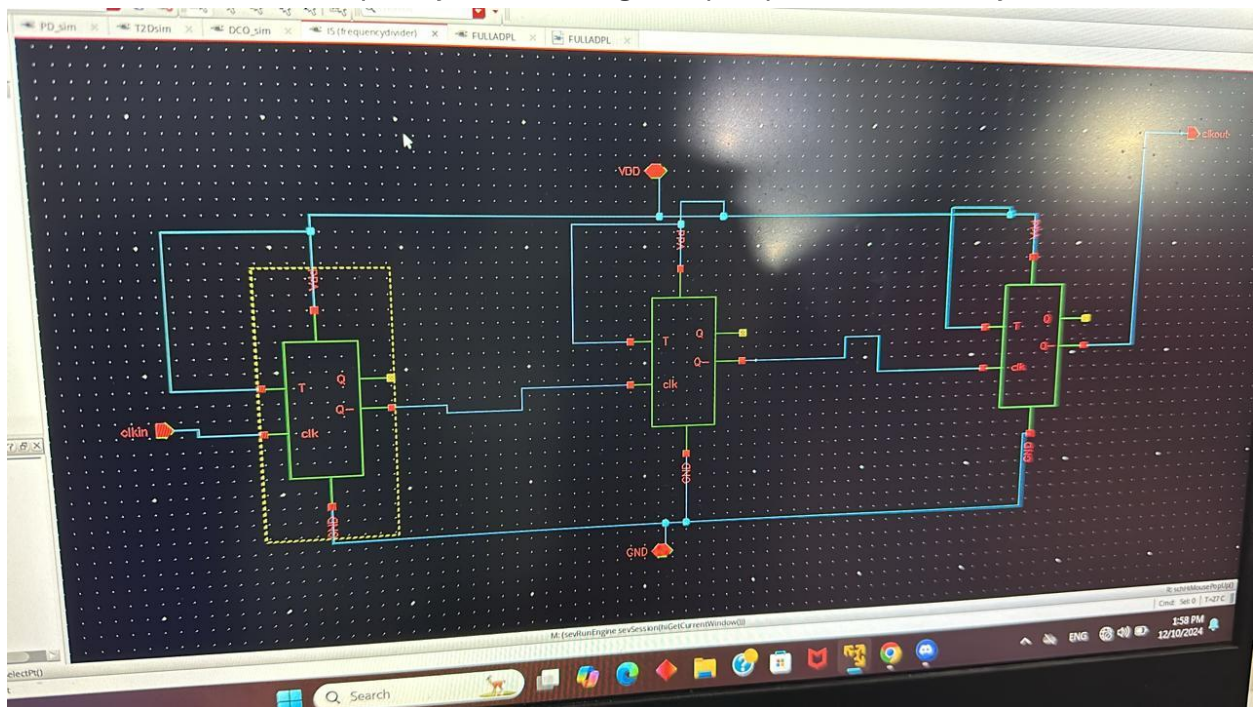
Digital control oscillator that produce oscillations using ring oscillators after that we got the input from 5 bit we get from loop filter and the output is the new CLK



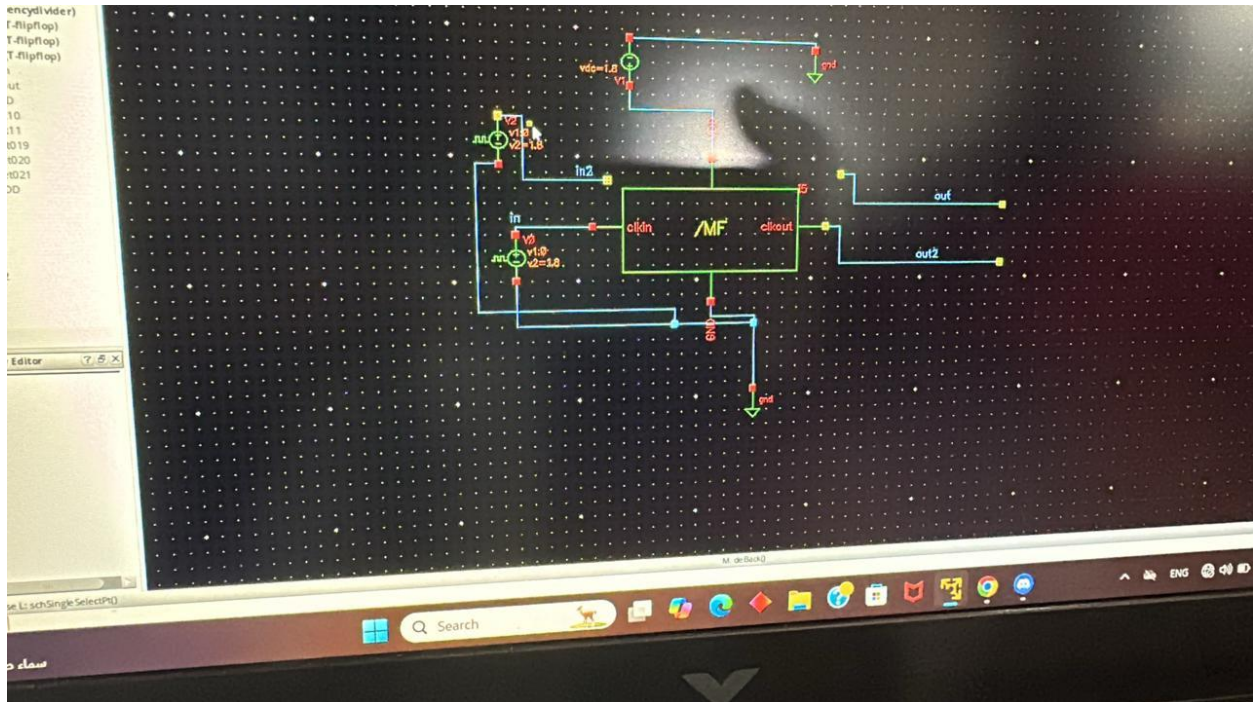
And we have simulated It separately



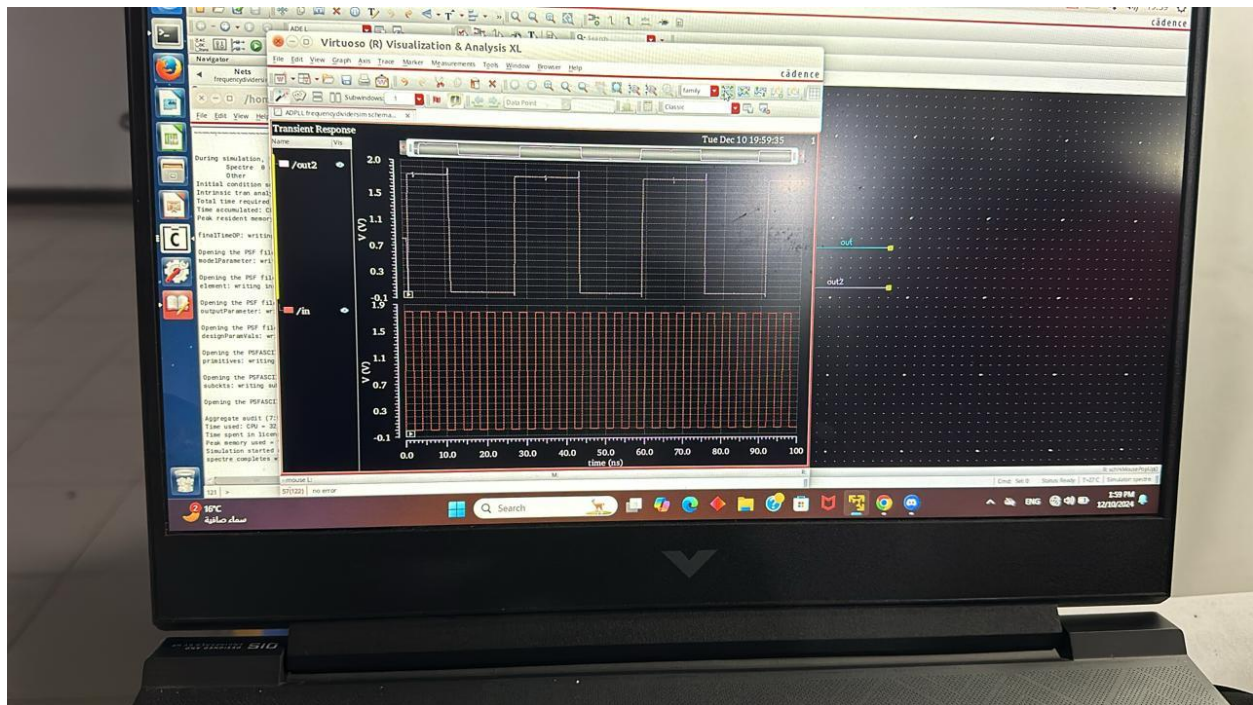
AT the end we use the frequency divider using 3 t flip flops to divide the by 8



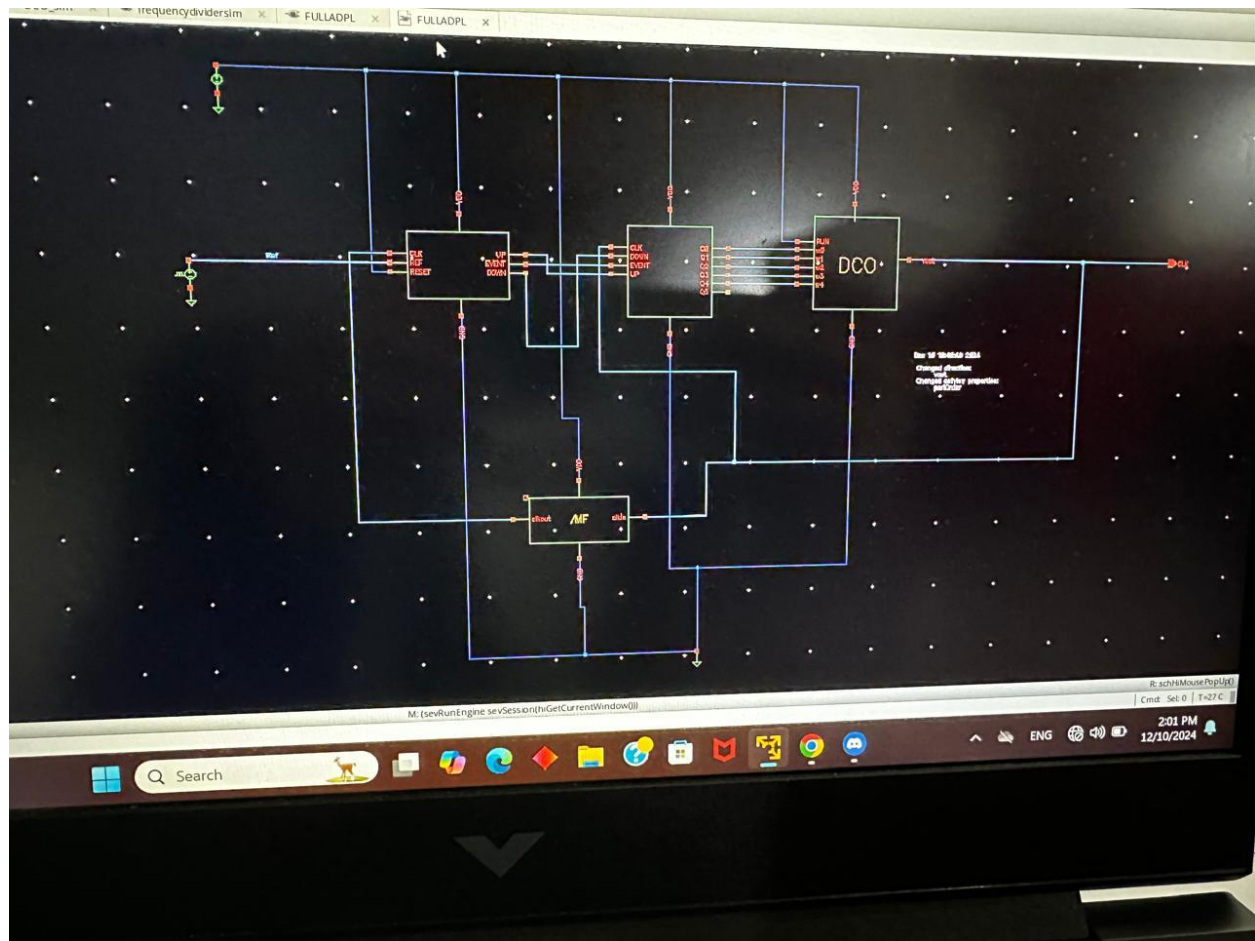
and we have simulated it separately



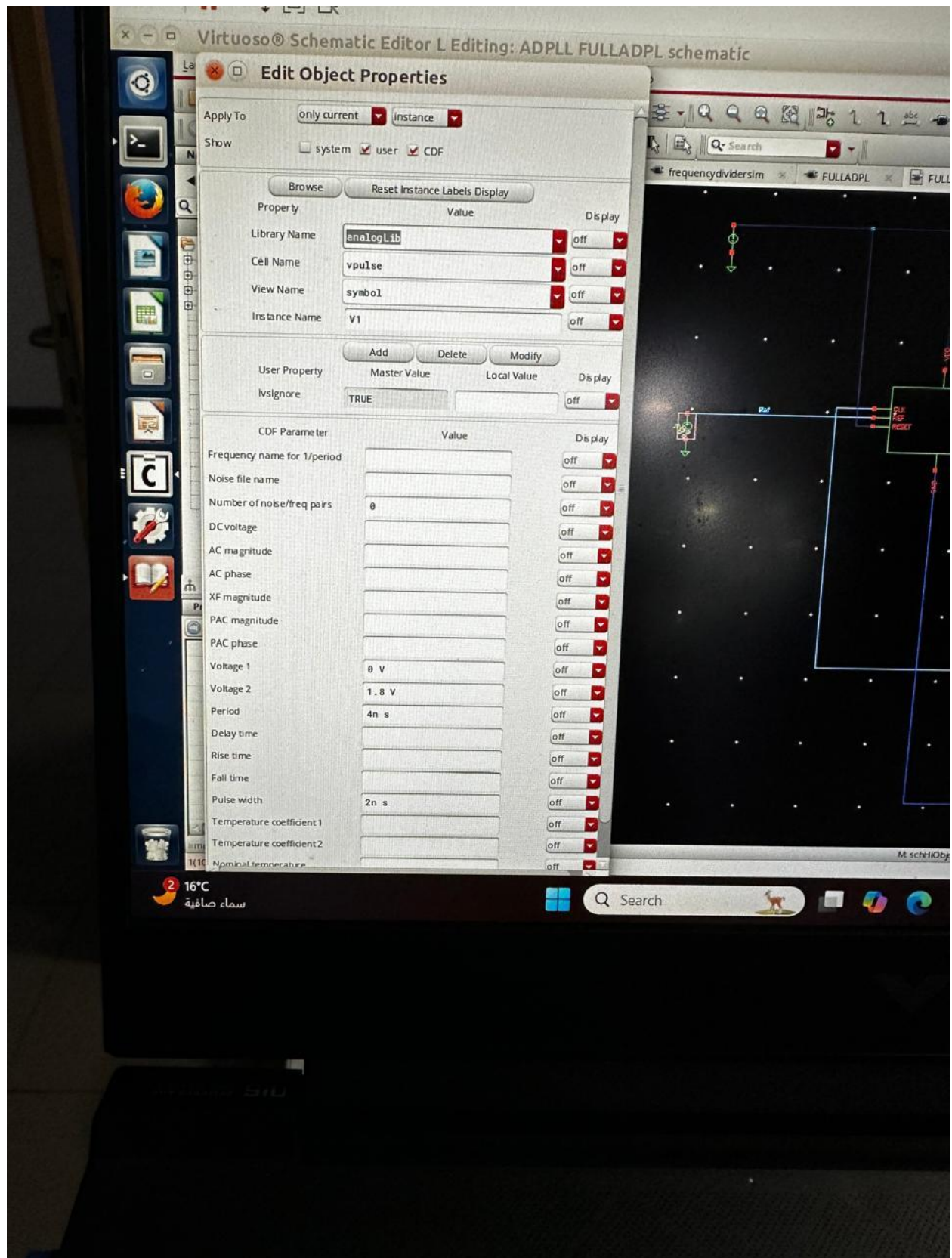
And the simulation is separately:



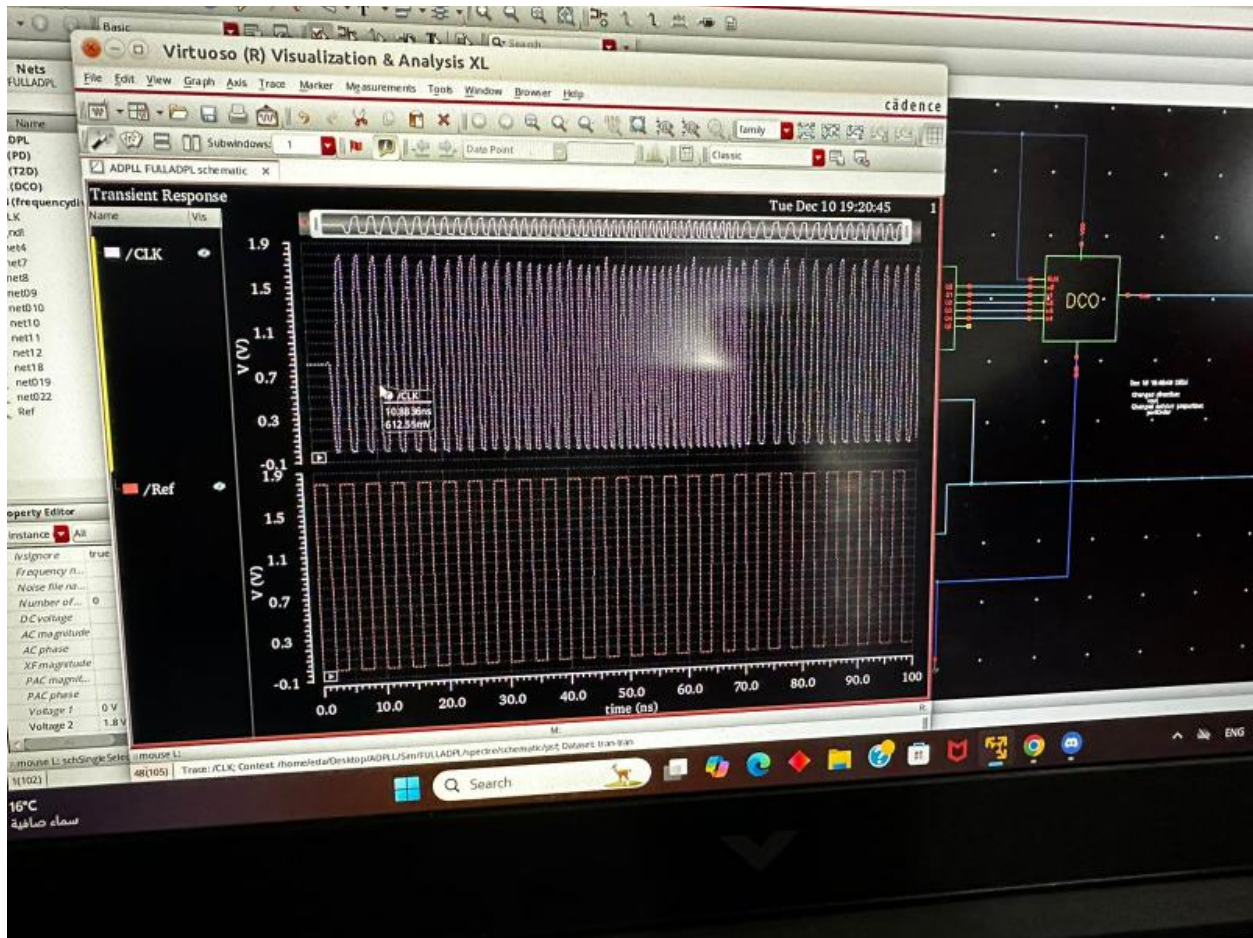
At the end we have connected all these components to design a full ADPLL



And we simulated it with REF signal :



And with vdd equal 1.8 volts and we simulated it to get :



By the end the CLK will be the same as the ref signal